



Data structure & Analysis of Algorithm
Mini Project Report
on

Electricity Board Management System

by

Nived Shenoy

Roll No: 160103245048

Dhairya Shah

Roll No: 16010324044

Course Co-ordinator

Prof.

Department of Electronics and Telecommunication Engineering
K. J. Somaiya School of Engineering
(Somaiya Vidyavihar University)

Contents

Algorithm

<https://github.com/CoderNived/Electricity-Board-Management>

Theory

The Electricity Board Management System is designed to automate electricity billing and consumer data management. Electricity usage is measured in kilowatt-hours (kWh), and bills are calculated based on the energy charge (unit rate $\times$ units consumed), fixed charge, and applicable taxes or surcharges. Data related to consumers, meter readings, and bills is stored using structured files or in-memory data structures, allowing efficient retrieval and updates while maintaining consistency. User authentication ensures secure access, and input validation prevents errors such as negative readings or duplicate entries. The system emphasizes modular design, dividing functions for easier maintenance and scalability, while following clear and user-friendly interaction flow. Efficient error handling enhances reliability, and though basic searches operate in $O(n)$ time, they can be optimized using faster data structures. Overall, the project integrates billing logic, secure data management, and structured software design to provide a simple yet effective electricity management solution.

Flowchart

https://www.canva.com/design/DAG1pcJwQfs/VcQtNjn5j8qfGsBnuKhq6Q/view?utm_content=DAG1pcJwQfs&utm_campaign=designshare&utm_medium=link2&utm_source=uniquelinks&utm_id=hoa2606c9f3

Presentation

https://www.canva.com/design/DAG1pmpmupo/4PLcgaDQ9oapSNM6HiC1Rg/view?utm_content=DAG1pmpmupo&utm_campaign=designshare&utm_medium=link2&utm_source=uniquelinks&utm_id=hof358c7941

Program code

<https://github.com/CoderNived/Electricity-Board-Management>

Future Enhancement/Development of the Project

The Electricity Board Management System can be improved by adding a graphical user interface (GUI) for better usability and integrating a database like MySQL for efficient data management. A consumer web portal with online payment options would enhance accessibility and convenience. Implementing slab-based billing, real-time notifications,

and data analytics can make billing more dynamic and informative. Strengthening security through encryption and role-based access, optimizing for scalability, and developing a mobile-friendly interface would further improve performance. Future integration with IoT-enabled smart meters could automate data collection and make the system more advanced and efficient.

Conclusion

In this project, we developed an Electricity Board Management System that allows management of consumers, recording of meter readings, and billing based on usage. The system ensures accurate computation of charges, maintains records, and offers basic CRUD functionality (Create, Read, Update, Delete) for consumers and bills. Through this work, I gained a deeper understanding of data structures, billing logic, file/database I/O, user interface design, and error handling.

However, the current implementation has limitations: it may not scale well for a large number of users, lacks a user-friendly GUI, and does not support online payment or real-time metering. The enhancements outlined earlier can make it more practical for real-world deployment.

Overall, this mini-project demonstrates a core backbone for an electricity management and billing system, which can be extended further in future work.